2025

Trial by Steel

GAME DESIGN DOCUMENT

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COMP 8800

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Status

The game is currently in the milestone 4 iteration. The core concept, scope, and overviews for player combat, boss archetypes, and major systems are complete, while the design for boss movesets and arenas is in progress. The player controls and moveset are complete with the full set of player movement and combat abilities implemented. Implemented the paladin boss which can chase and attack the player with jump, shield bash, and multiple melee combo attacks. Created an arena for the paladin boss fight. Implemented the beast boss which can slam the ground creating shock waves, throw a rock, leap at the player, and perform melee attacks. Created an arena for the beast boss fight. Implemented the sorcerer boss which can launch multiple magic projectiles, perform a meteor AOE attack, dash towards the player, and perform quick melee attacks. Created a final arena for the sorcerer boss fight. The player and the bosses can die and trigger the end of the level. All planned UI elements are present on the HUD. High scores for each boss are saved/loaded and displayed on a High Scores screen. Added an easy gameplay mode (selected through gameplay settings) that spawns a potion every 10 seconds on interaction that heals the player 50 health when interacted with. The project finished on schedule with no current blockers.

Tasks

Below is a link to the project board with all currently ready, in progress, and completed tasks.

Project Board Link: https://github.com/users/crystal-g77/projects/1/views/1

Concept Statement

Trial of Steel is a single-player boss-rush that delivers deep, expressive combat with a minimal control set: dodge, light attack, and heavy attack. Players build momentum through well-timed actions, chaining attacks and dodges to maximize damage and survive punishing, telegraphed boss encounters.

Genre(s)

Trial of Steel is a single-player action boss-rush game with elements of soulslike precision combat and momentum-based flow mechanics. The gameplay focuses on fast-paced, skillful duels against telegraphed bosses, rather than exploration or platforming. While rooted in action combat, the momentum system adds a layer of strategy and risk/reward pacing, distinguishing it from traditional hack-and-slash or pure Soulslike titles.

This combination creates a tight, replayable challenge, emphasizing readability, timing, and player mastery rather than expansive levels or complex resource management. The hybrid focus on action and flow-based mechanics ensures that each encounter feels deliberate, rewarding skillful play without overcomplicating the controls or systems.

Target Audience

Trial of Steel is aimed at single-player action gamers who enjoy challenging, skill-based combat and precise timing. The target players are likely fans of Soulslike games such as Dark Souls, Sekiro, or Hollow Knight, as well as players who appreciate fast-paced boss fights with clear telegraphs and risk/reward mechanics.

The intended audience is primarily teenagers and adults (ages 15+), comfortable with challenging gameplay and willing to invest time mastering combat systems. Players are expected to enjoy high-stakes duels, pattern recognition, and replayability rather than exploration-heavy or narrative-driven experiences.

The desired ESRB rating is T (Teen), reflecting fantasy violence, occasional gore from boss fights, and an emphasis on challenging but non-realistic combat.

Concept and Unique Selling Points

Concept

Trial of Steel is a single-player boss-rush game where players engage in fast, momentum-based combat against a series of challenging, telegraphed bosses. The core gameplay loop revolves around reading boss attack patterns, timing dodges, and chaining attacks to maintain momentum, rewarding skillful play and careful risk management. Players progress by facing increasingly difficult encounters, with each boss offering unique attack behaviors and arenas that challenge player adaptability. The game is set in a dark fantasy world, with minimalist, atmospheric arenas that focus attention on the combat itself. The visual style emphasizes dramatic lighting and impactful VFX to make every strike feel meaningful, while sound design and music heighten tension and emphasize the flow of combat.

Unique Selling Points (USPs)

- **Momentum-Based Combat**: Unlike typical Soulslike or hack-and-slash games, player actions build and maintain momentum, rewarding precise chains of attacks and dodges. This creates a fluid, satisfying combat rhythm.
- **Telegraphed, Skillful Boss Fights**: Each boss presents clear but challenging attack patterns, allowing players to anticipate and respond. The difficulty is high but fair, emphasizing readable design and mastery.
- **Focused Boss-Rush Structure**: By removing exploration and filler content, the game delivers a concise, intense, replayable experience centered entirely around combat encounters.
- **Atmospheric, Minimalist Arenas**: Arenas are designed to highlight the combat and boss telegraphs, using lighting, VFX, and sound to enhance tension and player immersion.

- **Accessible Controls, Deep Gameplay**: A minimal control set (dodge, light attack, heavy attack) keeps the game approachable, while depth emerges from timing, momentum, and strategy, rather than complicated mechanics.

Together, these features create a game that is easy to pick up, hard to master, and distinct from other boss-centric or Soulslike titles by emphasizing momentum-driven combat and highly readable, high-stakes encounters.

Player Experience and Game POV

In *Trial of Steel*, the player controls a lone warrior in a third-person perspective, facing a series of three punishing bosses in a dark fantasy world. The game takes place across distinct, atmospheric arenas designed to emphasize clarity and readability of combat. The experience is short but intense, with the full boss-rush lasting roughly 15–20 minutes. The player is shown the time it took to beat the boss and the max combo count they achieved after each boss along with a letter grade; this will increase replayability as the player will want to strive for a better letter grade.

The game grants the player the fantasy of mastering high-stakes, momentum-based combat, chaining attacks and dodges to outmaneuver formidable foes. Players are intended to feel tension, anticipation, and exhilaration, followed by satisfaction and accomplishment upon defeating each boss. Engagement is maintained through learning boss patterns, chaining attacks for momentum, and the sense of mastery gained from repeated practice.

Major Phases

1. **First Boss:** The player learns basic movement, attack timing, and momentum mechanics while immediately facing the first challenging boss in a third-person view, combining tutorial and skill test.
2. **Second Boss:** Difficulty ramps up, requiring the player to refine timing, read telegraphs carefully, and manage momentum under more complex attack patterns. This boss will attack much faster than the first boss increasing the reaction time necessary to dodge attacks to ramp up difficulty.
3. **Final Boss:** The climactic encounter tests mastery of all mechanics, delivering high-stakes tension and a strong sense of reward when overcome. This boss will have ranged attacks and be able to teleport around the arena making it so the player really has to learn the bosses move set in order to evade all attacks.

Key Moments

- **First Strike Against the Boss:** Experiencing the impact of a well-timed attack hooks the player and establishes combat feel.
- **Learning a Boss Pattern:** Recognizing and adapting to attack telegraphs creates satisfaction and skill progression.
- **Near-Defeat Moments:** Narrowly dodging attacks or surviving low-health situations generates tension and excitement.

- **Victory Over Final Boss:** Defeating the last boss delivers a strong sense of accomplishment and mastery, completing the player's arc.

Visual and Audio Style

Visual Style

Trial of Steel has a dark fantasy look with simple, atmospheric arenas that keep the focus on combat. Lighting is dramatic, with high-contrast shadows and bold silhouettes that make bosses and attack telegraphs easy to see. Bosses are designed to feel large, imposing, and memorable, using shape and movement to communicate threat and personality rather than relying on overly detailed textures. The overall visual style supports the game's tension and intensity while keeping the action readable.

Audio Style

The audio complements the visuals, with crisp, impactful sounds for attacks, dodges, and boss abilities to give clear feedback and emphasize momentum. Each boss has its own distinct audio cues for attacks, helping players anticipate patterns. Music is dynamic and tense, rising and falling with the flow of combat, while subtle environmental sounds enhance the dark, immersive atmosphere without distracting from gameplay.

References / Inspiration

- *Dark Souls* and *Sekiro* – minimalist, readable lighting and silhouette-driven design.
- *Hollow Knight* – dynamic music and responsive audio cues that reinforce gameplay.

Together, the visuals and audio make the player feel immersed, alert, and in control, highlighting the drama of every high-stakes boss fight.

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Concept Art

Player Character

The Paladin

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The Paladin's Arena

The Beast

Crystal Giesbrecht
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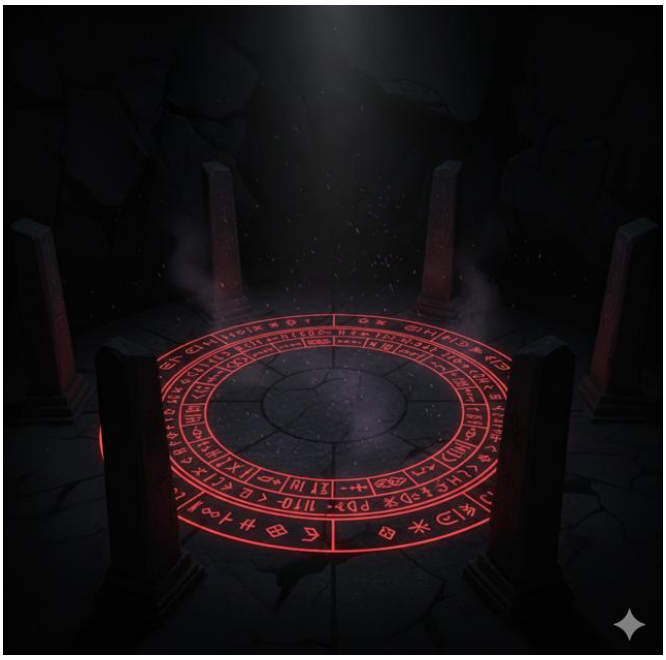
The Beast's Arena

The Sorcerer

The Sorcerer's Arena



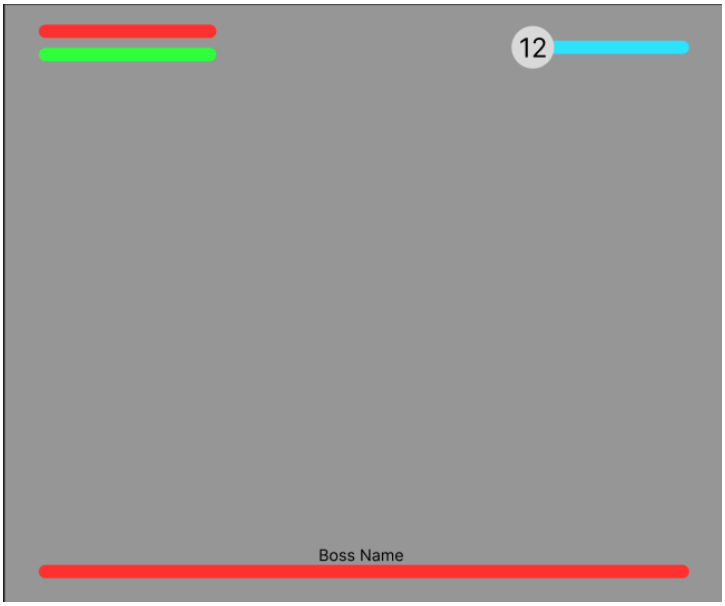
User Interface

Start Screen

Elements:

- Game Title
- Start Button

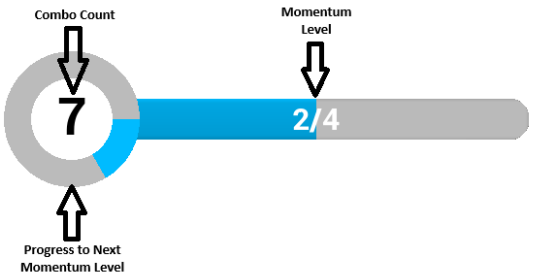
In-game HUD



Elements:

- Player health
- Player Stamina
- Player momentum
- Player combo counter
- Boss name
- Boss health

Momentum Meter



Elements

- Player momentum
- Player combo counter

Stat Screen

Elements:

- Heading
- Time it took to kill last boss
- Max combo achieved on last boss

Game World Fiction

Backstory Overview

The world of Trial of Steel is a shattered, dark fantasy realm, where powerful beings have seized control of fragmented arenas scattered across a cursed landscape. Legends speak of a lone warrior capable of challenging these bosses, restoring balance through skill and perseverance. The backstory sets the tone and atmosphere but does not directly affect gameplay; the focus is on skillful combat and mastery, not exploration or narrative branching.

World Organization

The game consists of three distinct arenas, each representing a boss encounter.

- **Arena 1 (Paladin)**: A ruined cathedral hall with wide open space and fallen statues. Minimal obstacles to keep focus on reading his long attacks.
- **Arena 2 (Beast)**: A broken courtyard overgrown with roots and rubble. Uneven ground adds tension but remains readable and fair.
- **Arena 3 (Sorcerer)**: A circular underground ritual chamber carved from black stone. Pulsing rune circles telegraph hazards, while a few pillars provide limited cover.

Game World Physics

Players move in a third-person perspective. Movement is grounded, focused on running, dodging, and positioning, no flying or complex traversal mechanics are included. Physics emphasize momentum-based combat, where chaining attacks and dodges builds flow and rewards precise timing with increased damage.

Player Progression Through Locations

Progression is linear, with players moving from the first arena through each subsequent boss arena. Each encounter is self-contained, promoting replayability and mastery of individual fights.

Competition

Trial of Steel shares similarities with several renowned action games known for their challenging combat and atmospheric design.

Key Competitors

Hollow Knight

Release Date: February 24, 2017
Developer: Team Cherry
Genre: Metroidvania
Revenue: Over $100 million
Description: An indie Metroidvania set in a vast, interconnected world, praised for its challenging combat and atmospheric design.

Sekiro: Shadows Die Twice

Release Date: March 22, 2019
Developer: FromSoftware
Genre: Action-Adventure
Sales: Over 10 million units
Description: An action-adventure game that combines stealth, exploration, and intense combat, set in a reimagined late 1500s Japan.

Dark Souls

Release Date: October 4, 2011
Developer: FromSoftware
Genre: Action RPG
Sales: Over 37 million units (series total)
Description: A challenging action RPG known for its deep lore, intricate world design, and punishing difficulty.

What Sets Trial of Steel Apart

Trial of Steel distinguishes itself by focusing on a concise, boss-rush experience with a momentum-based combat system that rewards skillful chaining of attacks and dodges. The game emphasizes clarity and readability in its minimalist arena designs, ensuring that players can focus entirely on mastering each encounter. With only three major bosses, the game offers a short but intense experience, catering to players seeking a challenging yet accessible action game.

Monetization

Trial of Steel will be a free-to-play single-player game, allowing anyone to experience the full boss-rush at no cost. Monetization, if any, will be cosmetic only, such as alternate player skins, visual effects, or non-gameplay-affecting boss variants.

This approach aligns with the design because:

- Skill-based progression is preserved: all players can access every arena and boss without paying.
- The core appeal, momentum-driven, high-stakes combat, remains untouched: maintaining fairness and challenge.
- Cosmetic options provide optional revenue: enhancing player expression without impacting gameplay or balance.

This model ensures that the game is accessible to a broad audience while still offering potential monetization through voluntary purchases.

Platform(s), Technology, and Scope

Platform

Trial of Steel will be developed for PC, supporting keyboard and mouse controls, and gamepads.

Technology

The game is being built in Unreal Engine 5, using 3D assets for characters, bosses, and arenas. Physics, animations, and visual effects leverage UE5's systems, including Nanite for detailed models and Lumen for dynamic lighting to enhance the dark fantasy atmosphere. Visual Studio 2022 is being used to edit C++ files.

Scope and Development

Team Size: Individual project (solo development).

Estimated Timeline: Approximately 8 months from planning to final milestone.

First Playable: Targeting Prototype 1 by October 12, featuring basic player movement and combat. Ability to lock on and attack a dummy enemy

Full Game Completion: Targeting Milestone 5 in April, with all three boss arenas completed and polished, including visuals, audio, and tutorial elements.

Major Risks:

- **Unreal Engine Learning Curve**: solo developer experience is limited, so mastering UE5 systems may slow progress.
- **Scope Creep**: ensuring the game remains focused on three bosses and core combat mechanics is critical.
- **Polish and Feedback**: limited ability to playtest widely may affect difficulty balance and combat feel.

Development Schedule and Milestones

Game Design Document (September 21)

Goal: Lock down the game concept and scope.

- Core pillars: simple but polished combat, telegraphed bosses, 3 bosses total (expandable to 4 if time allows).
- Combat system: light/heavy, dodge, jump, lock-on.
- Boss lineup: Paladin → Beast → Sorcerer.
- Art direction: minimal, using marketplace assets.

Deliverable*:* GDD that describes the mechanics, bosses, and scope.

Prototype (October 12)

Goal: Establish the core player mechanics and basic combat feel. Focus on movement, dodging, and attacking against a training dummy to validate combat.

- Playable player character with movement, jumps, and dodges
- Light and heavy attacks functional (full "ability" system) and consume stamina
- Player and enemy can both die to trigger lose/win condition
- Basic HUD feedback (health bar, stamina bar, and combo meter and counter)
- Placeholder arena environment
- Placeholder enemy

Deliverable*:* Simple combat prototype showing "game feel."

Milestone 1 (November 9)

Goal: Begin first boss encounter implementation, testing basic AI and attack telegraphs in a simple arena.

- Player vs boss 1 test scene
    - Initial arena layout
- Boss model
    - Basic boss attack patterns (telegraphed) with animations
- Placeholder VFX and audio for attacks

Deliverable: Playable first boss encounter prototype with basic combat and telegraphs.

Milestone 2: Alpha: MVP (December 9)

Goal: Complete first boss encounter, achieving Minimum Viable Product with core combat validated.

- Final arena design for first boss
- Complete first boss encounter
    - Full set of attacks and AI behavior
- Player health/damage feedback
- Intro animations added
- Tutorial/instructions integrated
- Playable loop with momentum combat validated

Deliverable: MVP showing first boss fully playable with core combat loop.

Milestone 3 (February 8)

Goal: Complete second boss encounter with unique mechanics and increased difficulty.

- Second boss model with full animations
- Unique attacks
- Arena for second boss
- VFX and audio specific to second boss

Deliverable: Second boss fully implemented and playable.

Milestone 4: Beta (March 15)

Goal: Complete final boss encounter and all gameplay content, including climactic arena and unique mechanics.

- Final boss model with full animations
- Climactic arena design
- Unique attacks
- Full VFX/audio integration

Deliverable: Final boss fully playable, completing all main gameplay content.

Milestone 5: Final (April 12)

Goal: Polish all game elements, integrate optional tutorial, and finalize player experience.

- Refinement of all boss attacks
- Bug fixes and performance optimization
- Polish visual and audio effects
- Final UI/UX improvements
- Add rank system (S - F) based on max combo count and time taken
- High scores for each boss saved/loaded and displayed
- Credits displayed
- Add easy mode with spawning consumable potions

Deliverable: Fully polished, playable game with refined combat, visuals, and audio.

Builds

Prototype

Below is a link to the Prototype build and a link to all completed tasks as of this build.

GitHub Release Link: https://github.com/crystal-g77/COMP8800-TrialbySteel/releases/tag/Prototype

Project Board Link: https://github.com/users/crystal-g77/projects/1/views/1?filterQuery=iteration%3APrototype

Milestone 1

Below is a link to the Milestone 1 build and a link to all completed tasks as of this build.

GitHub Release Link: https://github.com/crystal-g77/COMP8800-TrialbySteel/releases/tag/Milestone1

Project Board Link: https://github.com/users/crystal-g77/projects/1/views/1?filterQuery=iteration%3A%22Milestone+1%22

Milestone 2

Below is a link to the Milestone 2 build and a link to all completed tasks as of this build.

GitHub Release Link: https://github.com/crystal-g77/COMP8800-TrialbySteel/releases/tag/Milestone2

Project Board Link: https://github.com/users/crystal-g77/projects/1/views/1?filterQuery=iteration%3A%22Milestone+2%22

Milestone 3

Below is a link to the Milestone 3 build and a link to all completed tasks as of this build.

GitHub Release Link: https://github.com/crystal-g77/COMP8800-TrialbySteel/releases/tag/Milestone3

Project Board Link: https://github.com/users/crystal-g77/projects/1/views/1?filterQuery=iteration%3A%22Milestone+3%22

Milestone 4

Below is a link to the Milestone 4 build and a link to all completed tasks as of this build.

GitHub Release Link: https://github.com/crystal-g77/COMP8800-TrialbySteel/releases/tag/Milestone4

Project Board Link: https://github.com/users/crystal-g77/projects/1/views/1?filterQuery=iteration%3A%22Milestone+4%22

Milestone 5

Below is a link to the Milestone 5 build and a link to all completed tasks as of this build.

GitHub Release Link: https://github.com/crystal-g77/COMP8800-TrialbySteel/releases/tag/Milestone5

Project Board Link: https://github.com/users/crystal-g77/projects/1/views/1?filterQuery=iteration%3A%22Milestone+5%22

Objectives and Progression

Player Movement Through the Game

Players progress linearly through the game, moving from one boss arena to the next. The first arena combines tutorial elements and the first boss fight, teaching the player the fundamentals of movement, dodging, attacking, and the momentum-based combat system. Upon defeating a boss, the player is teleported directly to the next arena, maintaining momentum and tension. The second arena introduces increased complexity in attack patterns, requiring the player to refine their skills. The final arena presents the climactic boss, testing mastery of all mechanics learned so far.

Short-Term Goals

- Successfully land attacks and build momentum chains
- Dodge and respond to boss telegraphs effectively
- Survive individual boss encounters
- Learn and adapt to new attack patterns with each boss

Long-Term Goals

- Defeat all three bosses to complete the game
- Master the momentum-based combat system across all encounters
- Optimize combat strategies to achieve faster, more efficient victories and higher combo counts

How Goals Support Game Concept and Player Fantasy

The short-term and long-term goals reinforce the core fantasy of being a skilled warrior capable of reading and mastering complex attacks. The linear progression ensures that players focus entirely on combat mastery, while the increasing difficulty of bosses provides satisfying challenge and skill growth. Each goal aligns with the dark fantasy aesthetic and the momentum-driven combat loop, rewarding timing, precision, and strategic decision-making.

Core Loops

Primary Gameplay Loop

The core loop revolves around momentum-based combat against bosses:

- **Observe**: The player watches boss attack telegraphs and arena cues.
- **React**: The player dodges, positions, and chains light/heavy attacks to maintain momentum.
- **Execute**: Successful attacks build momentum, deal damage, and create visual/audio feedback that reinforces skillful play.
- **Adapt**: Players learn and refine strategies as bosses' attack patterns evolve, leading to mastery over time.

This loop is engaging because it rewards skill, timing, and observation, creating a satisfying feedback cycle where players immediately see the impact of their actions.

Support for Player Goals

- Short-term goals (surviving and landing attacks) are reinforced with every successful dodge or strike.
- Long-term goals (defeating all bosses and mastering combat) are supported by repeated exposure to the loop with increasing difficulty and complexity.

Emergent Results

- Players may develop unique strategies or combos tailored to specific bosses.
- Replayability emerges naturally as players attempt faster or more efficient clears and higher combo counts.
- Optional cosmetic goals in a F2P model could emerge if skins or effects are tied to performance milestones, though they do not affect core gameplay.

Game Systems and Features

Movement System

Intent (Why): Movement is the foundation of the player's survival and combat flow. Dodging, spacing, and repositioning must feel fluid and responsive to make boss encounters fair and engaging.

Moment-to-Moment Gameplay:

- Use mouse move (keyboard + mouse) or right stick (controller) to control camera.
- Use WASD or arrow keys (keyboard + mouse) or left stick (controller) to mov.
- Use Space (keyboard + mouse) or A button (controller) to jump.
- Use CTR (keyboard + mouse) or B Button (controller) to dodge.
- Dodging consumes stamina.

UI Description:

- Player Health bar (top left), stamina bar (below health bar).

Technical Notes:

- Root motion animations for dodges to ensure smooth transitions.
- Implemented with Character + Animation Blueprints.
- Movement responsiveness tuned through acceleration/deceleration curves in Blueprint/C++.

Core Combat System

Intent (Why): Combat is the heart of the game. It needs to feel responsive, readable, and satisfying.

Moment-to-Moment Gameplay:

- Use left click (keyboard + mouse) or Right Bumper (controller) to light attack.
- Use right click (keyboard + mouse) or Right Trigger (controller) to heavy attack.
- Light and heavy attacks consume stamina.
- Player observes enemy telegraph → reacts with dodge or jump. Use dodging with I-frames to evade attacks. Use jumping to go over certain attacks.
- Finds openings → lands light or heavy attacks.
- Failing to dodge/jump results in a small stagger where player cannot perform another action.

UI Description:

- Player Health bar (top left), stamina bar (below health bar).

Visual Effects:

- Hit sparks and red vignette on player damage.

Technical Notes:

- Damage implemented with built-in Apply Point Damage and Event Point Damage, passing in the character that is being hit.
- Hit detection implemented with traces using weapon collision volume.

Momentum System

Intent (Why): Adds a risk–reward layer to combat, rewarding consistent dodging and attacking with a momentum combo that boosts damage.

Moment-to-Moment Gameplay:

- Successful attack or dodge increases the Combo counter.
- If hit, or if the combo timer runs out, the chain breaks.
- Combo count translates into a damage multiplier (Momentum) based on steps. There will be 5 steps requiring an increasing combo count to progress to the next 1. Each step increase is a +10% damage multiplier.
    - Step 1 → Step 2: Total combo count 2 (2 from previous)
    - Step 2 → Step 3: Total combo count 6 (4 from previous)

    - Step 3 → Step 4: Total combo count 12 (6 from previous)
    - Step 4 → Step 5: Total combo count 20 (8 from previous)

UI Description:

- Player Momentum bar and Combo count on the top right
- Bar fills and changes color as momentum increases.

Visual Effects:

- Player weapon glows or trails during active momentum. This is more of a stretch goal.
- Intensified sparks when momentum is high. This is more of a stretch goal.

Technical Notes:

- Blueprint variables tracks momentum state and timer.

AI Boss Encounters (3 Unique Fights)

Intent (Why): Boss fights teach players to read attacks, adapt, and master combat mechanics in high-stakes scenarios. Bosses must feel intelligent and deliberate without being overwhelming. Their AI is designed around distance-based decision-making and cooldown pacing, ensuring attacks feel fair, telegraphed, and punishable.

Moment-to-Moment Gameplay:

- Distance Check: Boss constantly measures player distance.
    - Close range → favors melee combos, shield bashes, or short range attacks.
    - Mid range → uses gap closers like charges, leaps, or pounces.
    - Long range → executes ranged attacks or approaches cautiously.
- Cooldown Tracking: Each attack type has an internal cooldown, preventing spam and creating varied rhythms.
- Telegraph + Commit: Once an attack is chosen, boss clearly telegraphs it (animation, sound, VFX) and commits fully, no mid-attack canceling.
- Recovery Window: After an attack, bosses leave intentional punish windows for the player to strike back.
- Idle/Movement Filler: When no action is valid (e.g., cooldowns active, player out of range), boss repositions or performs a "threaten" animation to keep tension.

UI Description:

- Boss health bar (bottom center).

Visual Effects:

- Telegraphed attacks.

Technical Notes:

- Boss AI via state tree component.
- Telegraphs tied to animation notifies.

Arena Transition System

Intent (Why): Seamlessly moves players between encounters, maintaining momentum and tension, and provides feedback on player performance.

Moment-to-Moment Gameplay:

- Player defeats a boss → displays stat screen → arena fades/teleports to the next boss arena.
- Transition maintains immersion without breaking combat flow.

UI Description:

- Stat screen displays time taken to beat boss and highest momentum multiplier achieved.

Technical Notes:

- Teleport system triggers on boss defeat after stat screen display.
- Telegraphs tied to animation notifies.

Minimal Narrative Framing

Intent (Why): Provides context and fantasy without distracting from core combat.

Moment-to-Moment Gameplay:

- Short narrative prompts at the start of each arena or boss as well as a short introduction animation
- Environmental storytelling through arena design (ruins, ritual chambers).

UI Description:

- Subtle text overlays or cut-in animations introducing arena/boss.

Technical Notes:

- Arena Blueprints reference narrative assets to trigger text or cut-ins at the start of each encounter.
- Trigger events in the arena (e.g., OnBossSpawn, OnArenaEnter) drive narrative playback.

Game Objects

The Samurai

Purpose: The player character.

Visual Style: 2-handed sword wielding, melee character with cloth armor.

Combat Style: Fast sword attacks.

Key Mechanics:

- Light and heavy sword attacks
- Dodge
- Jump

The Paladin

Purpose: Serves as the tutorial-style first boss. Introduces players to core combat flow: spacing, dodging, and punishing openings.

Visual Style: Towering, armored knight with a broad shield and massive axe. Heavy plate armor, slightly battered, gives weight to every swing.

Combat Style: Slow, deliberate melee with wide sweeps and shield bashes. Telegraphs are clear but punishing if ignored.

Key Mechanics:

- Sweeping axe arcs (teaches dodge timing + spacing)
- Shield bash stun (punishes greedy attacks)
- Occasional ground slam (introduces area-of-effect attacks)

The Beast

Purpose: Escalates difficulty by testing reaction time and precision dodging. Builds on lessons from The Paladin but adds speed and aggression.

Visual Style: Hulking quadrupedal beast. Long claws and exaggerated muscle mass emphasize raw aggression.

Combat Style: Relentless lunges, claw swipes, and short charge attacks. Keeps constant pressure on the player.

Key Mechanics:

- Rapid claw combos (tests dodge roll timing)
- Charge-and-pounce attacks (forces repositioning)

The Sorcerer

Purpose: Final test of movement, awareness, and punishing small openings. Pushes the player to read the battlefield and manage hazards.

Visual Style: Tall, gaunt figure with flowing robes and a hood that obscure much of the body. Robes are layered, floating unnaturally. Wields small 1-handed sword.

Combat Style: AOE rune circles, magic projectiles, and teleport repositioning. Attacks punish poor awareness and force deliberate dodges.

Key Mechanics:

- Meteor AOE spell (tests awareness + movement)
- Magic projectile volleys (forces varied dodging)
- Teleport reposition (keeps pressure unpredictable)
- Quick sword attacks

Localization

The game will launch English-only for the MVP. Future updates may include EFIGS (English, French, Italian, German, Spanish) depending on player demand and resources. All in-game text will be stored in a UE5 String Tables, allowing runtime localization. Strings will use placeholders for dynamic values (e.g., {Boss1_Name}) to be replaced at runtime. The format will be [Context_Identifier].

Technical Documentation

Major Technical Areas and Risks

- Core Combat & Momentum System: Ensuring responsive input, proper hit detection, and consistent momentum tracking. Risk: timing bugs or desync between animation and collision.
- Boss AI: Implementing clear telegraphs and predictable-but-challenging behavior. Risk: AI state machine complexity may cause bugs or exploitable patterns.
- Arena Transitions: Smooth teleportation and stat screen display without breaking immersion. Risk: potential for camera or collision issues.
- Performance: Maintaining 60+ FPS with VFX, particle effects, and multiple boss arenas. Risk: high particle count could drop performance on lower-end PCs.

Target Hardware and Requirements/Assumptions

- **Platform**: PC (Windows 10+)
- **Assumptions:**
    - Keyboard + mouse or controller support.
    - Minimum GPU: NVIDIA GTX 1050 / equivalent.
    - CPU: Quad-core 2.5 GHz+
    - RAM: 8 GB minimum

Artificial Intelligence and Autonomous Systems:

- **Boss AI**: State machine with attack phases, telegraph timers, and behavior trees.
- **Procedural Systems**: None required for MVP (arenas are prebuilt).
- **Pathfinding**: Simple NavMesh for boss positioning within arenas.
- **Economy**: None required for MVP (F2P monetization is cosmetic only).

Ideas and Expansions

- Add monetary cosmetics (player model skins, weapon model skins)
- Add phases to boss encounters (add moves, increase speed of existing moves)

- Add different player character archetypes (magic user, archer)
- Add more bosses

Questions

Resolved

- How complex should boss AI navigation be within arenas? Will a NavMesh be sufficient for limited movement arenas?
    - NavMesh will be sufficient for limited movement in conjunction with AI State Tree
- How to structure Blueprints for modular arenas, bosses, and combat systems while avoiding circular dependencies.
    - Individual bosses all inherit from a common base AI character blueprint
    - Each boss blueprint contains components that manage health, abilities, and a state tree
        - A boss's abilities are determined by an array of ability enums along with data for that ability passed to the Ability component
    - State tree contains tasks for moving and all the different abilities that the boss has (these tasks can be shared across boss state trees)
- Deciding between Blueprint heavy or C++ heavy implementation for performance and maintainability.
    - Going with more of a Blueprint-heavy implementation, reserving C++ usage to functions that are far more difficult or impossible to implement in Blueprints (UE has some features and functions that are only available in C++)
- How to centralize momentum and combat logic to avoid duplicated code across bosses.
    - Using actor components to represent things like health, stamina, and combo count with the actor itself as well as dispatchers to communicate between the different components.
    - Using an ability system so that implementing new attacks scales and at the same time allows for specific sequences of events per ability
- Is it better to use State Trees or Behaviour Trees for boss AI?
    - State Tree pros
        - Allows for more control in determining which state to transition to next
    - State Tree cons
        - Seems to be no runtime view of a State Tree, unlike a Behavior Tree
    - Can overcome con by using debug prints to determine states and tasks
    - Going with State Trees
- How does terrain editor work in Unreal Engine 5? Should the terrain editor or pre-built assets be used to build the arenas? Or some combination of the two?
    - Most likely pre-built assets will suffice. Shouldn't really be a need for custom terrain.

Change Log

Date	Version	Author	Description
September 21, 2025	1.0	Crystal	• Initial draft complete
October 16, 2025	2.0	Crystal	• Updated Status section with details about prototype being completed • Updated second and third boss details in Major Phases section of Player Experience and Game POV • Added features that were implemented in prototype that were unplanned in Development Schedule and Milestones section • Shuffled features and details a little bit between Milestones 1 and 2 in Development Schedule and Milestones section • Updated Questions section to include a Resolved section for the questions that I have determined an answer to as well as a brief description of the answer
October 19, 2025	2.1	Crystal	• Added new unresolved questions to Questions section • Slightly changed wording in Emergent Results section of Core Loops • Updated Paladin in Concept Art Section • Updated Beast Visual and Combat Style in GameObjects section • Slightly changed wording in Long-Term Goals section of Objectives and Progression • Updated Localization section
November 13, 2025	3.0	Crystal	• Provided answers to some unresolved questions in Questions section, moving them to resolved • Updated features in Milestone 2 in Development Schedule and Milestones section (replaced feature that was done ahead of time with new one)
November 16, 2025	3.1	Crystal	• Moved location of Combo count and Momentum bar to the top right in User Interface in Concept Art section • Updated UI Description in Momentum System in Game Systems and Features

			section to match new location of Combo count and Momentum bar • Updated Technical Notes in Core Combat System in Game Systems and Features section
November 19, 2025	4.0	Crystal	• Updated Status section • Added Builds section with Github links to the releases and project board tasks per build • Added Tasks to Status section with link to project board
November 25, 2025	4.1	Crystal	• Updated Paladin's arena concept art in Concept Art section • Moved Question from unresolved to resolved in Questions section
November 30, 2025	4.2	Crystal	• Added annotated Momentum meter to Concept Art section under In-Game HUD
February 1, 2026	5.0	Crystal	• Added dates to Milestones 3,4, and 5 in Development Schedule and Milestones section • Added Milestone 2 to Builds section
February 6, 2026	5.1	Crystal	• Cleaned up wording for tasks in Milestones 3 and 4 in Development Schedule and Milestones section • Updated Status section • Added Milestone 3 to Builds section
March 12, 2026	6.0	Crystal	• Added Samurai (player character) section to GameObjects • Updated Status section • Added Milestone 4 to builds section
April 10, 2026	7.0	Crystal	• Updated Status section • Added Milestone 5 to builds section • Added high scores, credits screen, pause screen, and easy mode to Milestone 5 in Development Schedule and Milestones section • Updated title font to match in game